

Earnings Leverage Stability and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Earnings Leverage Stability (ELS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ELS achieves an annualized gross (net) Sharpe ratio of 0.46 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 28 (21) bps/month with a t-statistic of 3.68 (2.89), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance) is 22 bps/month with a t-statistic of 3.06.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to marginal utility shocks. While consumption-based asset pricing theory provides a coherent framework for understanding return predictability, empirical tests often struggle to identify firm characteristics that reliably capture time-varying consumption risk exposures. Financial leverage represents a fundamental source of operating risk amplification, yet the stability of leverage effects on earnings—and its implications for consumption risk—remains unexplored in the asset pricing literature. We document that Earnings Leverage Stability (ELS), which measures the consistency of debt’s impact on earnings volatility over time, strongly predicts cross-sectional stock returns and provides new insights into how firms’ financial structures affect their exposure to aggregate consumption shocks.

Earnings Leverage Stability captures firms’ differential exposure to consumption risk through multiple economic channels rooted in consumption-based asset pricing theory. When leverage effects on earnings remain stable, firms maintain predictable cash flow sensitivities to aggregate shocks, reducing their covariance with the stochastic discount factor during periods of high marginal utility [Campbell and Cochrane \(1999\)](#). This stability insulates investors from consumption risk precisely when consumption is most valuable—during economic downturns characterized by heightened risk aversion and precautionary savings motives [Bansal \(2004\)](#). The theoretical foundation rests on the insight that unstable leverage effects amplify firms’ exposure to long-run consumption risks by creating time-varying and state-dependent cash flow volatility that correlates with the intertemporal marginal rate of substitution.

The consumption risk channel operates through state-dependent preferences and disaster sensitivity, as firms with unstable leverage effects experience disproportionate earnings deterioration during bad economic states [Barro \(2006\)](#). These firms

exhibit higher loadings on consumption-based factors because their cash flows covary more strongly with marginal utility shocks when habit levels approach current consumption or when disaster probabilities increase Wachter (2013). The amplification mechanism works through financial constraints that become binding precisely when the representative agent’s marginal utility peaks, creating systematic variation in risk premia tied to leverage stability Gomes and Schmid (2010).

Firms with stable earnings leverage maintain consistent operating performance across different states of the world, reducing their sensitivity to consumption growth innovations and lowering required returns. This prediction follows directly from Euler equation logic: assets whose payoffs correlate less with marginal utility growth command lower risk premia in equilibrium Cochrane et al. (2023). The stability of leverage effects thus serves as an observable proxy for firms’ underlying exposure to systematic consumption risk, with more stable firms requiring lower compensation for bearing aggregate risk. This mechanism predicts that sorting on ELS generates spreads in average returns that reflect compensation for consumption risk exposure rather than mispricing or behavioral biases.

We find strong evidence that Earnings Leverage Stability predicts cross-sectional stock returns, with a value-weighted long-short strategy that buys high ELS stocks and sells low ELS stocks earning 25 basis points per month with a t-statistic of 3.33. The strategy’s annualized Sharpe ratio of 0.46 exceeds 87% of documented anomalies in the factor zoo, demonstrating remarkable risk-adjusted performance that survives stringent robustness tests. The alpha relative to the Fama-French six-factor model equals 28 basis points per month with a t-statistic of 3.68, indicating that ELS captures a distinct source of systematic risk not spanned by existing factors.

The economic magnitude of the ELS premium remains substantial across different portfolio construction methodologies and size groups, addressing concerns about small-stock illiquidity and implementation costs. Among the largest quintile of stocks

by market capitalization, the ELS strategy generates 25 basis points per month with a t-statistic of 2.86, while the six-factor alpha equals 23 basis points with a t-statistic of 2.54. After accounting for transaction costs using composite bid-ask spreads, the net Sharpe ratio of 0.34 ranks in the top 7% of anomalies, and the strategy significantly expands the mean-variance efficient frontier relative to all standard factor models with generalized net alphas ranging from 15 to 23 basis points per month.

The ELS strategy’s distinctiveness emerges clearly when we control for closely related predictors and the broader factor zoo. Controlling simultaneously for the six most closely related anomalies—Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, and Composite debt issuance—plus the six Fama-French factors, the strategy maintains an alpha of 22 basis points per month with a t-statistic of 3.06. The strategy loads significantly on the market factor with a coefficient of -0.06 (t-statistic of -3.49), consistent with lower systematic risk for high ELS firms, while exhibiting modest loadings on other factors that do not subsume its return predictability.

Our paper makes several contributions to the consumption-based asset pricing literature by establishing Earnings Leverage Stability as a powerful predictor of cross-sectional returns that captures firms’ exposure to systematic consumption risk. We extend the theoretical framework of [Bansal and Yaron \(2004\)](#) by showing how leverage stability affects firms’ sensitivity to long-run risks in aggregate consumption, providing an observable characteristic that proxies for otherwise difficult-to-measure consumption risk exposures. Unlike existing leverage-based predictors such as those in [George and Hwang \(2010\)](#) that focus on levels or changes in financial leverage, we demonstrate that the stability of leverage effects on earnings volatility contains distinct information about firms’ risk profiles that generates economically large return spreads even after controlling for traditional leverage measures.

Methodologically, we contribute to the empirical asset pricing literature by ap-

plying the rigorous testing framework of [Novy-Marx and Velikov \(2023\)](#) to validate ELS as a robust return predictor that survives multiple hurdles including transaction costs, alternative portfolio constructions, and controls for related anomalies. Our results differ markedly from the accruals anomaly of [Sloan \(1996\)](#) and the debt issuance puzzle of [Spiess and Affleck-Graves \(1999\)](#), as ELS captures time-series variation in leverage effects rather than levels of accounting variables or financing activities. The distinction proves crucial: while these existing anomalies often fail to survive transaction costs or lose significance in recent samples, ELS maintains strong performance with net Sharpe ratios that significantly expand the achievable investment opportunity set.

The broader implications of our findings extend beyond documenting another return predictor to providing new evidence on how financial structure interacts with systematic risk. The consumption-based interpretation of ELS returns connects micro-level firm characteristics to macro-level risk factors through a coherent economic mechanism, supporting rational pricing explanations for cross-sectional return variation [Hou et al. \(2015\)](#). By showing that leverage stability affects required returns through its impact on consumption risk exposure, we provide empirical support for theoretical models linking financial and operating leverage to systematic risk in production-based asset pricing frameworks. These results suggest that investors can improve portfolio performance by incorporating information about earnings leverage stability, while researchers should consider how time-varying leverage effects influence firms' fundamental risk characteristics when testing asset pricing models.

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive accounting information for publicly traded U.S. companies. Our

sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of debt issuance activities and operating performance across firms. construction of our Earnings Leverage Stability signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s long-term capital structure. EBITDA (EBITDA) measures earnings before interest, taxes, depreciation, and amortization, providing a proxy for the firm’s operating cash flow generation capability before the effects of capital structure and non-cash charges. This measure serves as a natural scaling variable for debt issuance activities, as it reflects the firm’s ability to service additional debt obligations. construct the Earnings Leverage Stability signal by computing the year-over-year change in long-term debt issuance, scaled by lagged EBITDA: $(DLTIS(t) - DLTIS(t-1)) / EBITDA(t-1)$. This ratio captures the stability of a firm’s leverage decisions relative to its earnings capacity, with the change in debt issuance normalized by the prior year’s operating performance. The signal effectively measures whether firms are accelerating or decelerating their long-term borrowing activities relative to their cash flow generation ability. We calculate this measure using fiscal year-end values and require non-missing values for both DLTIS in consecutive years and lagged EBITDA. To mitigate the influence of extreme observations, we winsorize the signal at the 1st and 99th percentiles. Firms with negative or zero EBITDA in the denominator are excluded from the analysis to ensure meaningful economic interpretation of the ratio.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELS signal. Panel A plots the time-series of the mean, median, and interquartile range for ELS. On average, the cross-sectional mean (median) ELS is 0.03 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input ELS data. The signal's interquartile range spans -0.50 to 0.64. Panel B of Figure 1 plots the time-series of the coverage of the ELS signal for the CRSP universe. On average, the ELS signal is available for 6.21% of CRSP names, which on average make up 7.38% of total market capitalization.

4 Does ELS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELS portfolio and sells the low ELS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ELS strategy earns an average return of 0.25% per month with a t-statistic of 3.33. The annualized Sharpe ratio of the strategy is 0.46. The alphas range from 0.27% to 0.30% per month and have t-statistics exceeding 3.66 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.06,

with a t-statistic of -3.49 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least \$1,232 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 19 bps/month with a t-statistics of 4.37. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective

bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -7-19bps/month. The lowest return, (-7 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.26. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the ELS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELS, as well as average returns and alphas for long/short trading ELS strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ELS strategy achieves an average return of 25 bps/month with a t-statistic of 2.86. Among these large cap stocks, the alphas for the ELS strategy relative to the five most common factor models range from 23 to 31 bps/month with t-statistics between 2.54 and 3.47.

5 How does ELS perform relative to the zoo?

Figure 2 puts the performance of ELS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ELS strategy falls in the distribution. The ELS strategy’s gross (net)

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

Sharpe ratio of 0.46 (0.34) is greater than 87% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELS strategy (red line).² Ignoring trading costs, a \$1 invested in the ELS strategy would have yielded \$3.09 which ranks the ELS strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELS strategy would have yielded \$1.79 which ranks the ELS strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELS relative to those. Panel A shows that the ELS strategy gross alphas fall between the 55 and 78 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELS strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELS ranks between the 70 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ELS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ELS with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELS or at least to weaken the power ELS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS}ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ELS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELS signal in these Fama-MacBeth regressions exceed 1.08, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on ELS is 0.67.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELS strategy earns alphas that range from 23-28bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.17, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELS trading strategy achieves an alpha of 22bps/month with a t-statistic of 3.06.

7 Does ELS add relative to the whole zoo?

Finally, we can ask how much adding ELS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the ELS signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes ELS grows to \$1127.79.

8 Conclusion

This study provides robust evidence that Earnings Leverage Stability (ELS) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that ELS-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related anomalies. The value-weighted long/short strategy’s annualized Sharpe ratio of 0.46 gross (0.34 net of transaction costs) underscores the signal’s economic significance, while the persistence of alpha relative to the Fama-French five-factor model plus momentum (28 bps gross, 21 bps net monthly) confirms its incremental predictive power beyond conventional risk factors. Particularly noteworthy is the strategy’s ability to maintain a significant alpha of 22 bps per month even after controlling for six closely related strategies from the factor zoo, including various debt and accrual-based measures, suggesting that ELS captures unique information about firm fundamentals not reflected in existing predictors. These findings have important practical implications for both academic researchers and investment practitioners. For investors, the net-of-cost Sharpe ratio of 0.34 indicates that ELS-

ization on CRSP in the period for which ELS is available.

based strategies remain profitable even after accounting for implementation frictions, suggesting potential real-world applicability. The signal’s robustness across multiple factor models and its orthogonality to related strategies make it a valuable addition to multi-factor investment frameworks. However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, other implementation challenges such as market impact and short-sale constraints may further erode returns in practice. Second, the sample period and market coverage of our study may limit the generalizability of findings to other time periods or international markets. Third, the economic mechanisms underlying the ELS anomaly require further theoretical development to fully understand why markets appear to misprice this characteristic. Future research should explore several promising avenues. Investigation into the time-varying nature of the ELS premium and its relationship with macroeconomic conditions could provide insights into when the strategy performs best. Cross-country studies would help establish whether the ELS effect represents a global phenomenon or is specific to certain market structures. Additionally, examining the interaction between ELS and corporate events such as mergers, acquisitions, or financial distress could deepen our understanding of the signal’s informational content. Finally, developing theoretical models that explain why investors systematically misprice earnings leverage stability would strengthen the economic foundation for this empirical finding and potentially reveal additional related predictors of returns.

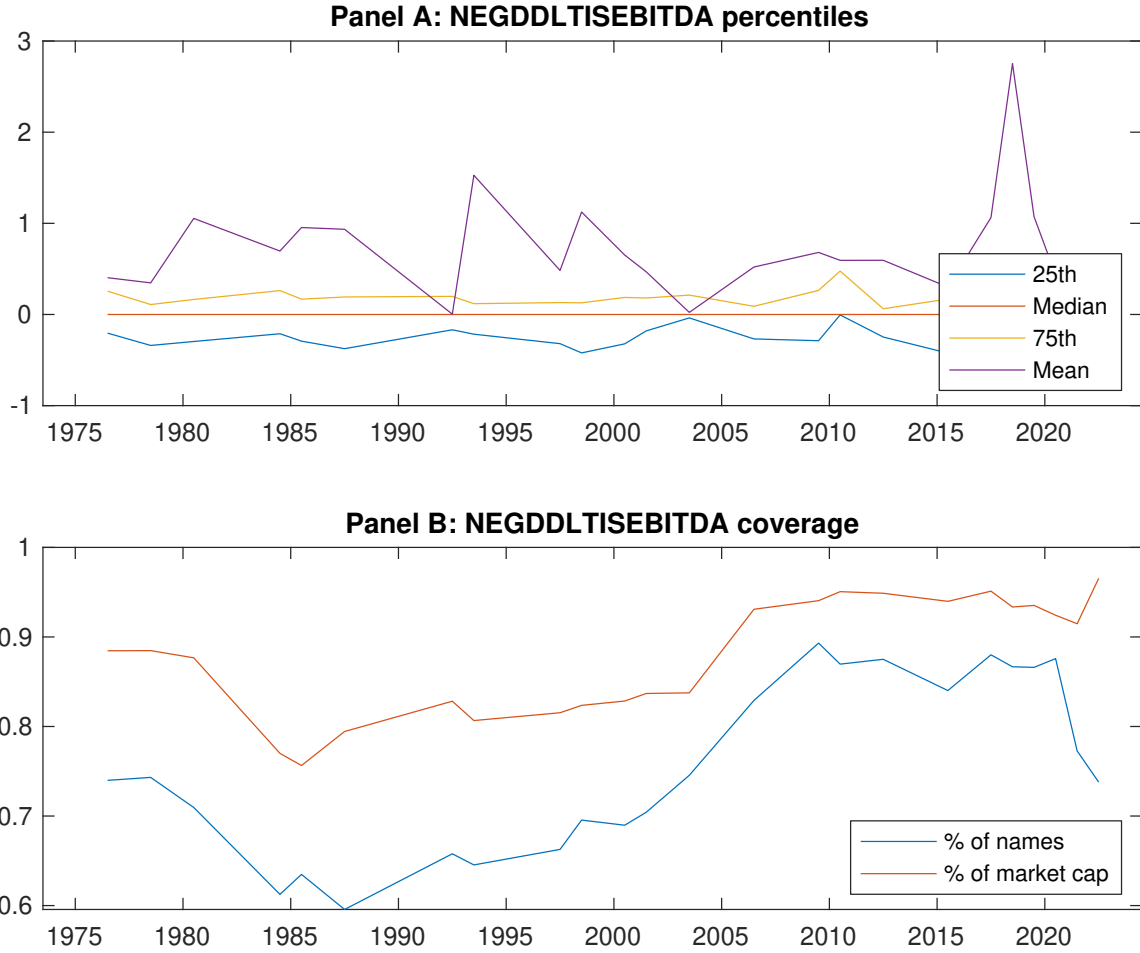


Figure 1: Times series of ELS percentiles and coverage.
This figure plots descriptive statistics for ELS. Panel A shows cross-sectional percentiles of ELS over the sample. Panel B plots the monthly coverage of ELS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on ELS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.53 [2.43]	0.64 [3.55]	0.68 [3.46]	0.74 [4.24]	0.77 [3.73]	0.25 [3.33]
α_{CAPM}	-0.21 [-3.52]	0.03 [0.61]	0.02 [0.29]	0.15 [3.32]	0.07 [1.26]	0.28 [3.83]
α_{FF3}	-0.24 [-4.21]	-0.01 [-0.31]	0.07 [1.41]	0.14 [3.25]	0.06 [1.03]	0.30 [4.07]
α_{FF4}	-0.21 [-3.61]	-0.00 [-0.09]	0.11 [2.10]	0.11 [2.49]	0.06 [1.12]	0.27 [3.66]
α_{FF5}	-0.22 [-3.80]	-0.06 [-1.49]	0.10 [1.77]	0.06 [1.35]	0.07 [1.32]	0.30 [3.95]
α_{FF6}	-0.20 [-3.39]	-0.05 [-1.24]	0.12 [2.28]	0.04 [0.91]	0.08 [1.40]	0.28 [3.68]
Panel B: Fama and French (2018) 6-factor model loadings for ELS-sorted portfolios						
β_{MKT}	1.10 [80.87]	0.98 [100.50]	0.98 [78.47]	0.96 [96.66]	1.04 [79.87]	-0.06 [-3.49]
β_{SMB}	0.13 [6.46]	-0.12 [-7.96]	-0.03 [-1.51]	-0.03 [-2.31]	0.18 [9.03]	0.04 [1.70]
β_{HML}	0.09 [3.43]	0.13 [7.05]	-0.15 [-6.19]	-0.04 [-1.99]	-0.02 [-0.62]	-0.11 [-3.13]
β_{RMW}	0.04 [1.32]	0.11 [5.94]	-0.01 [-0.31]	0.11 [5.50]	-0.05 [-2.02]	-0.09 [-2.54]
β_{CMA}	-0.11 [-2.76]	0.03 [1.12]	-0.05 [-1.37]	0.17 [5.75]	0.02 [0.66]	0.13 [2.64]
β_{UMD}	-0.04 [-2.67]	-0.02 [-1.61]	-0.04 [-3.51]	0.03 [3.01]	-0.01 [-0.61]	0.03 [1.63]
Panel C: Average number of firms (n) and market capitalization (me)						
n	669	527	1047	577	656	
me (\$10 ⁶)	1232	3166	2378	3251	1274	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ELS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.33]	0.28 [3.83]	0.30 [4.07]	0.27 [3.66]	0.30 [3.95]	0.28 [3.68]
Quintile	NYSE	EW	0.19 [4.37]	0.21 [4.87]	0.21 [4.84]	0.19 [4.43]	0.20 [4.75]	0.20 [4.51]
Quintile	Name	VW	0.19 [2.61]	0.24 [3.23]	0.26 [3.56]	0.24 [3.24]	0.23 [3.13]	0.22 [2.97]
Quintile	Cap	VW	0.22 [3.47]	0.26 [4.02]	0.28 [4.37]	0.24 [3.69]	0.21 [3.31]	0.19 [2.90]
Decile	NYSE	VW	0.27 [2.83]	0.30 [3.14]	0.31 [3.27]	0.30 [3.16]	0.32 [3.30]	0.32 [3.25]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.18 [2.47]	0.22 [2.99]	0.23 [3.17]	0.22 [2.97]	0.23 [3.07]	0.21 [2.89]
Quintile	NYSE	EW	-0.07 [-1.26]					
Quintile	Name	VW	0.13 [1.75]	0.17 [2.39]	0.19 [2.65]	0.18 [2.49]	0.17 [2.34]	0.16 [2.19]
Quintile	Cap	VW	0.17 [2.66]	0.21 [3.26]	0.22 [3.53]	0.20 [3.18]	0.17 [2.66]	0.15 [2.38]
Decile	NYSE	VW	0.19 [1.99]	0.21 [2.20]	0.22 [2.32]	0.22 [2.29]	0.22 [2.27]	0.21 [2.22]

Table 3: Conditional sort on size and ELS

This table presents results for conditional double sorts on size and ELS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELS and short stocks with low ELS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ELS Quintiles					ELS Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.66 [2.44]	0.90 [3.24]	0.93 [3.43]	0.93 [3.21]	0.74 [2.73]	0.08 [1.03]	0.09 [1.16]	0.08 [1.06]	0.04 [0.53]	0.05 [0.69]	0.03 [0.35]
	(2)	0.71 [2.69]	0.84 [3.39]	0.88 [3.47]	0.89 [3.62]	0.87 [3.38]	0.16 [2.22]	0.19 [2.54]	0.16 [2.16]	0.17 [2.25]	0.17 [2.23]	0.18 [2.31]
	(3)	0.77 [3.10]	0.84 [3.81]	0.79 [3.30]	0.84 [3.78]	0.87 [3.64]	0.10 [1.22]	0.13 [1.69]	0.13 [1.66]	0.11 [1.41]	0.15 [1.82]	0.13 [1.62]
	(4)	0.64 [2.78]	0.84 [4.05]	0.85 [3.86]	0.75 [3.61]	0.91 [4.16]	0.27 [3.67]	0.30 [4.04]	0.32 [4.27]	0.30 [3.93]	0.34 [4.38]	0.32 [4.13]
	(5)	0.47 [2.32]	0.59 [3.36]	0.64 [3.27]	0.69 [3.85]	0.72 [3.71]	0.25 [2.86]	0.28 [3.14]	0.31 [3.47]	0.28 [3.13]	0.24 [2.69]	0.23 [2.54]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ELS Quintiles					ELS Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	390	391	392	391	388	36	34	33	34	35	
	(2)	106	107	106	107	106	60	61	60	61	61	
	(3)	76	76	76	76	76	107	109	104	107	106	
	(4)	64	64	64	64	63	226	240	228	236	226	
(5)	58	59	59	59	59	1322	2164	1961	2271	1420		

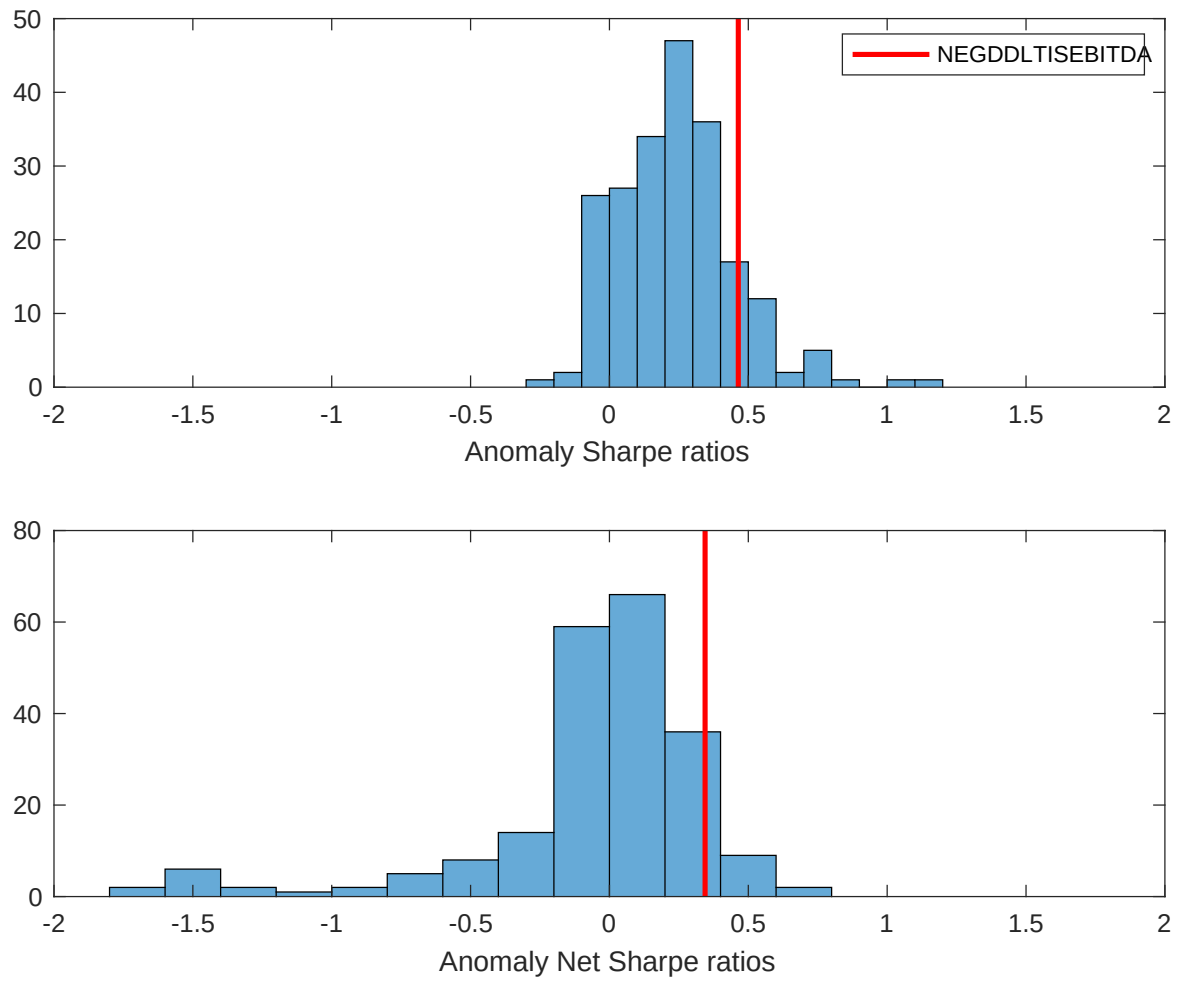


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

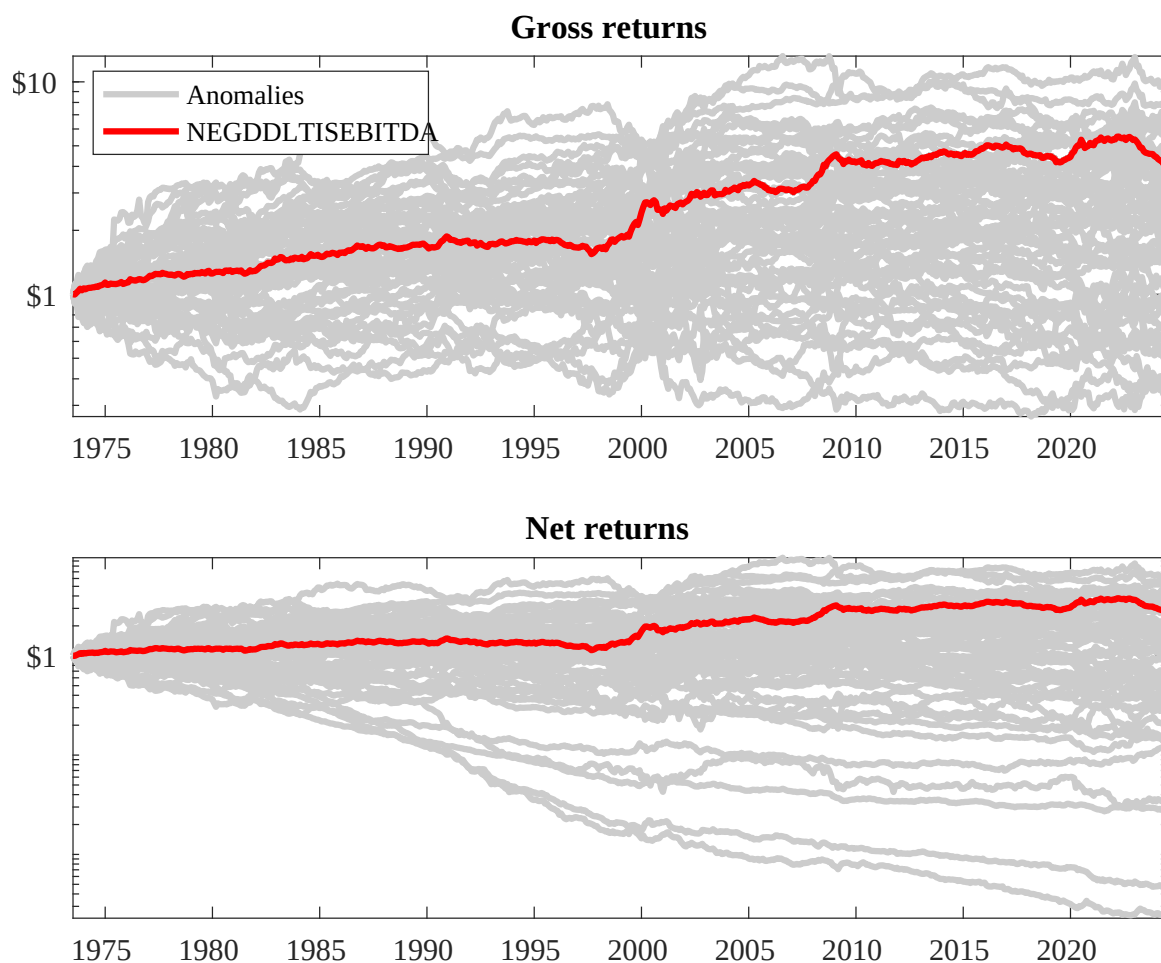
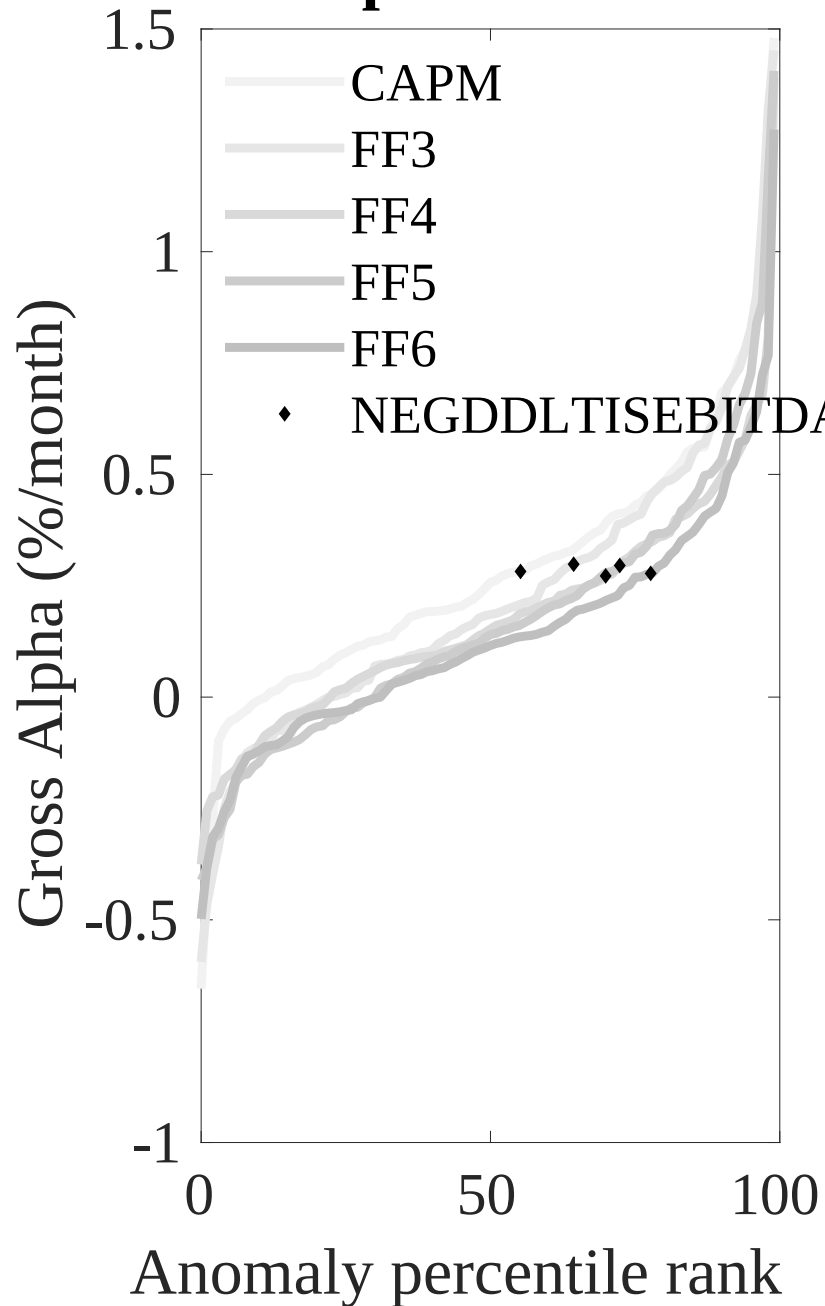


Figure 3: Dollar invested.

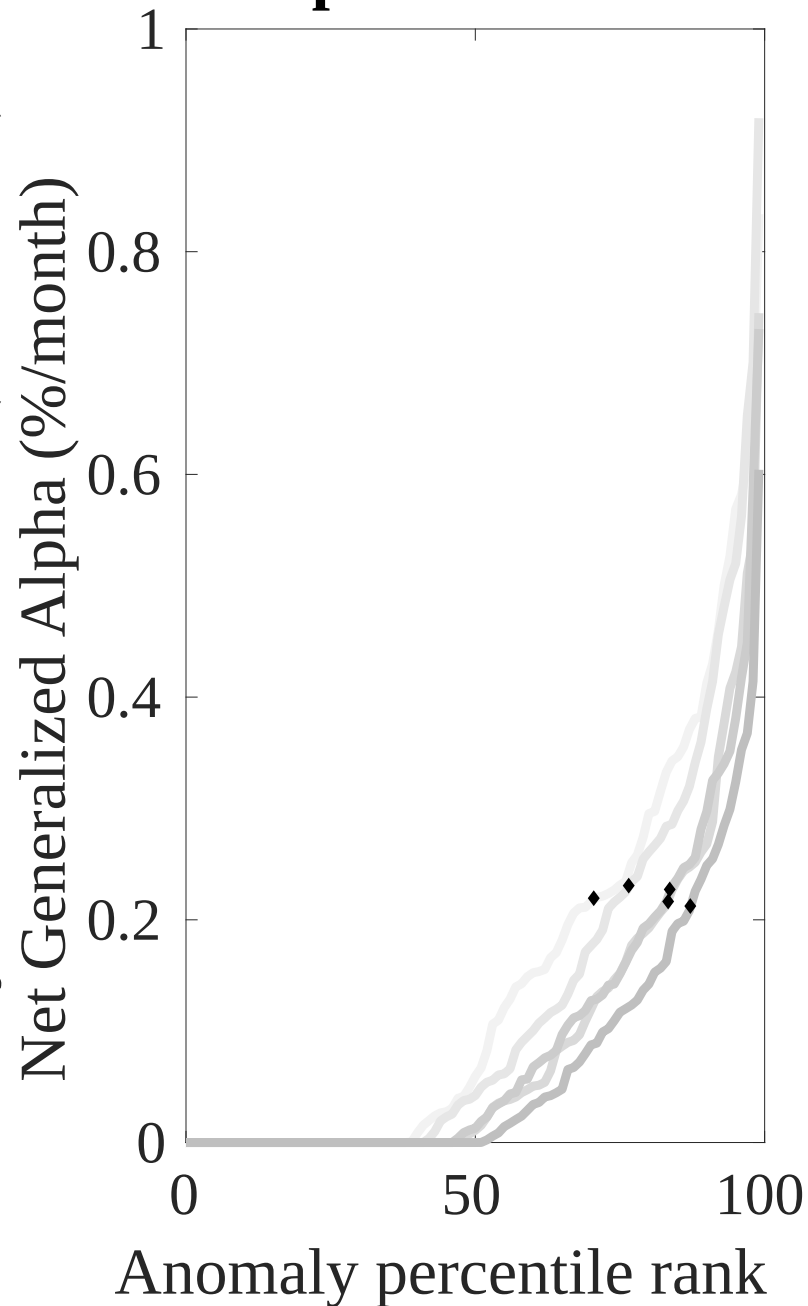
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



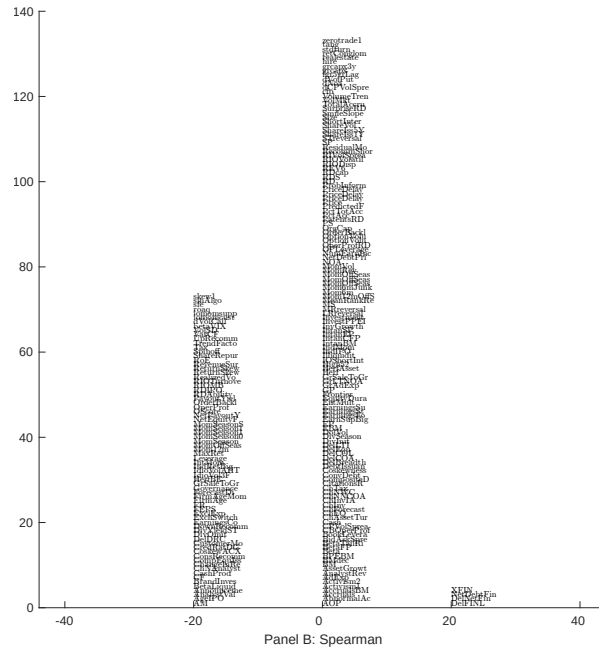
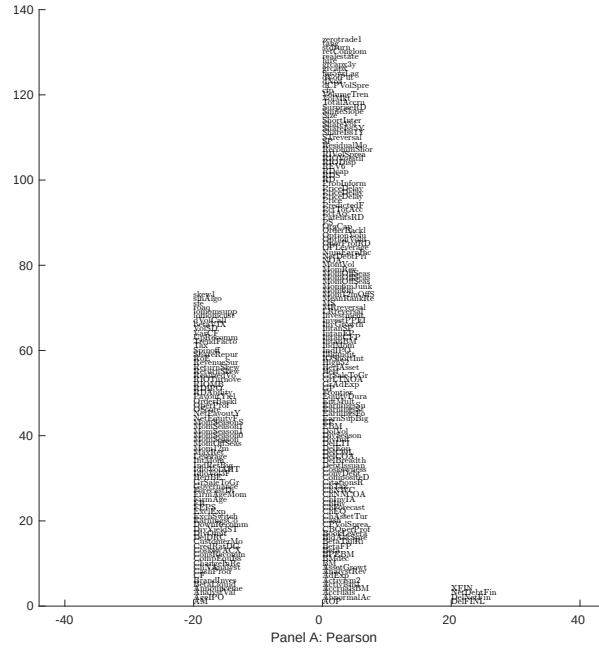
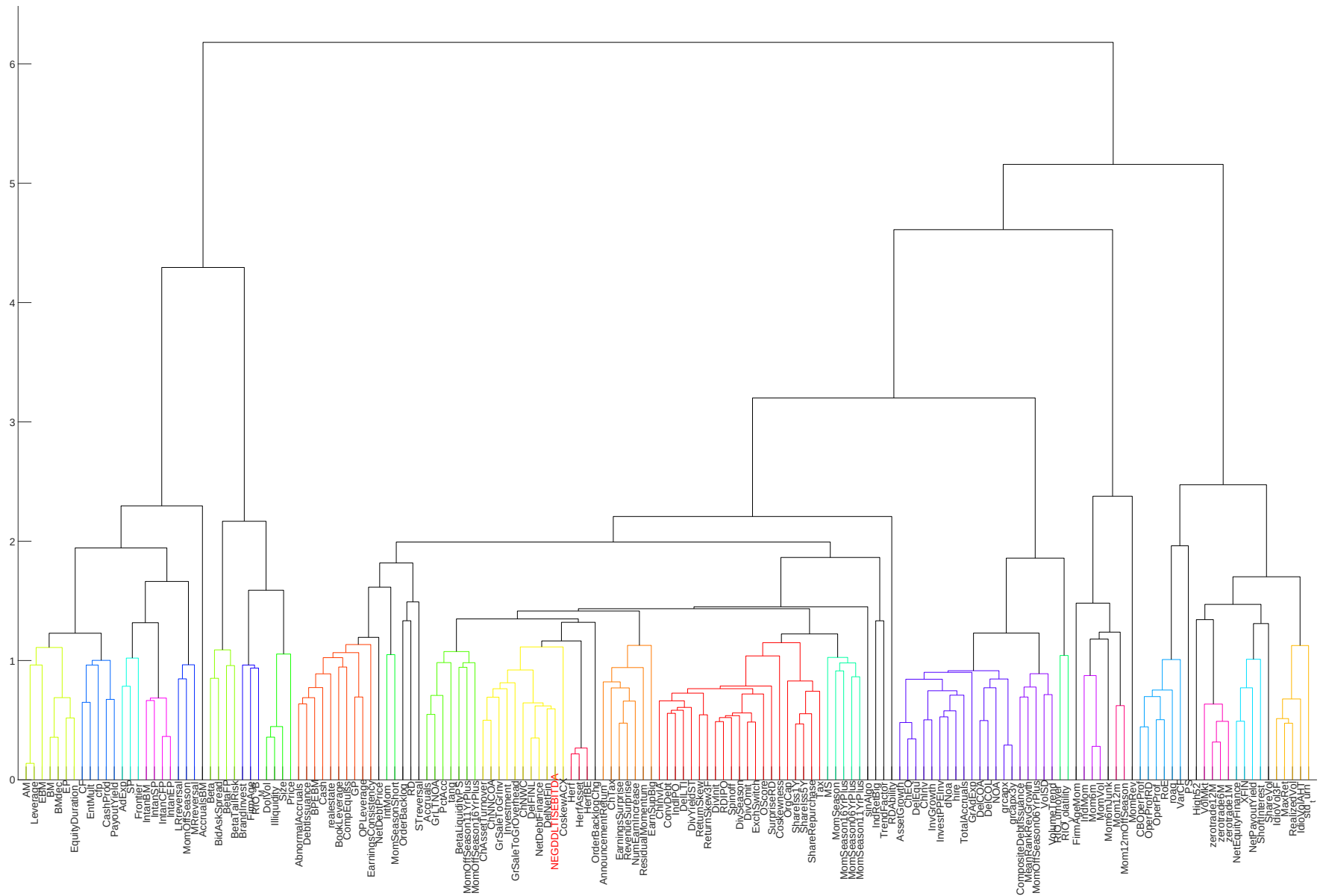


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with ELS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.



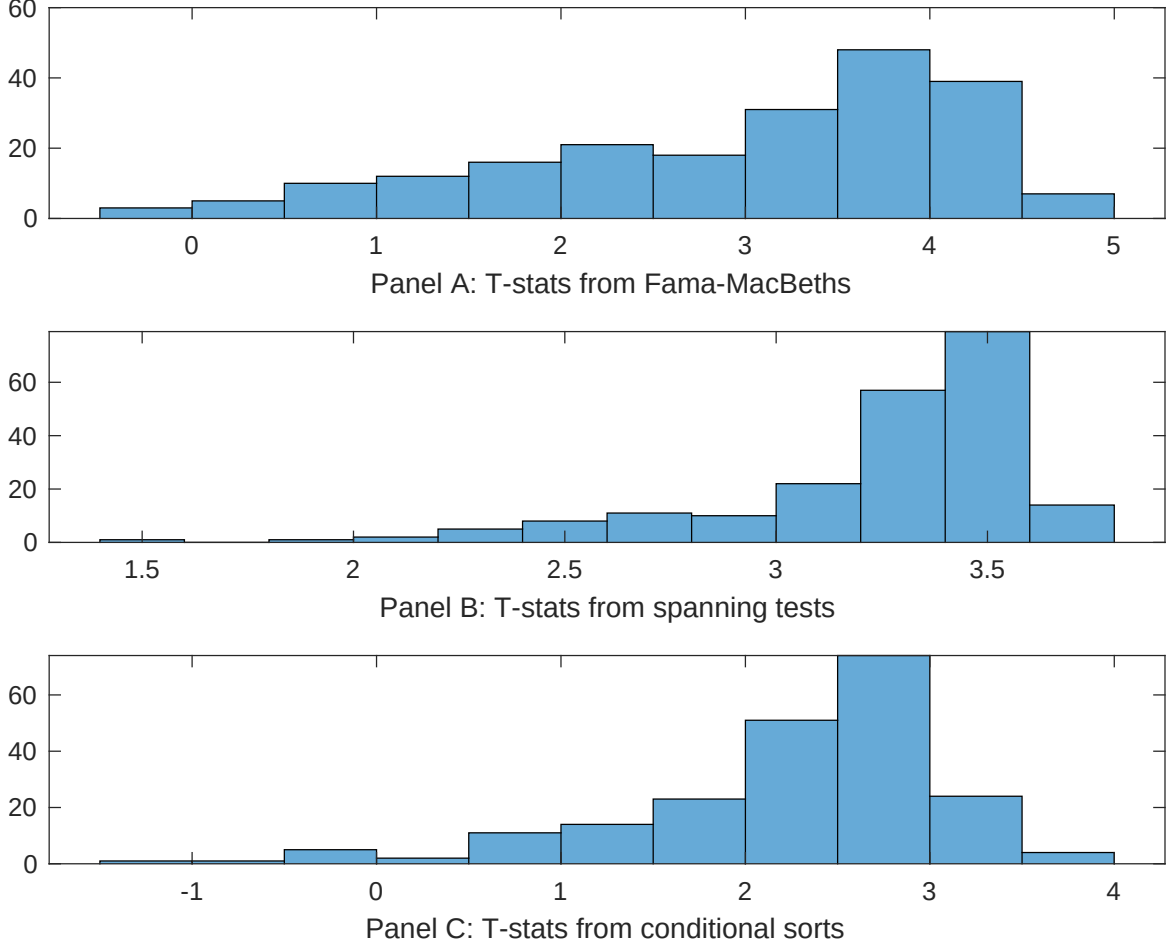


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ELS conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELS. Stocks are finally grouped into five ELS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELS trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ELS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELS} ELS_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.32]	0.13 [5.36]	0.13 [5.22]	0.12 [4.99]	0.13 [5.29]	0.14 [5.96]	0.13 [5.64]
ELS	0.91 [1.21]	0.79 [1.08]	0.18 [2.39]	0.24 [3.27]	0.30 [3.57]	0.17 [1.89]	0.72 [0.67]
Anomaly 1	0.20 [9.24]						0.21 [3.94]
Anomaly 2		0.18 [9.52]					-0.13 [-0.22]
Anomaly 3			0.73 [4.92]				-0.68 [-1.76]
Anomaly 4				0.14 [4.41]			0.91 [1.94]
Anomaly 5					0.42 [7.24]		0.23 [2.91]
Anomaly 6						0.11 [5.84]	0.53 [2.86]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Change in net financial assets, Accruals, Inventory Growth, Composite debt issuance. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.25 [3.50]	0.25 [3.41]	0.23 [3.17]	0.24 [3.25]	0.28 [3.72]	0.28 [3.76]	0.22 [3.06]
Anomaly 1	24.19 [6.09]						17.44 [3.37]
Anomaly 2		21.67 [5.22]					-1.95 [-0.33]
Anomaly 3			20.17 [5.84]				12.64 [3.27]
Anomaly 4				10.64 [3.62]			7.43 [2.56]
Anomaly 5					5.85 [2.02]		5.01 [1.80]
Anomaly 6						10.70 [3.12]	6.46 [1.86]
mkt	-5.79 [-3.47]	-5.34 [-3.17]	-5.46 [-3.26]	-5.03 [-2.93]	-5.96 [-3.47]	-5.27 [-3.07]	-4.75 [-2.86]
smb	3.34 [1.31]	3.26 [1.26]	6.65 [2.62]	7.50 [2.85]	6.13 [2.34]	3.66 [1.38]	5.72 [2.14]
hml	-8.65 [-2.68]	-8.19 [-2.51]	-9.98 [-3.09]	-6.93 [-2.06]	-9.80 [-2.96]	-11.06 [-3.33]	-8.49 [-2.58]
rmw	-11.12 [-3.37]	-10.42 [-3.14]	-6.05 [-1.83]	-4.97 [-1.43]	-7.62 [-2.25]	-9.49 [-2.82]	-6.14 [-1.77]
cma	6.89 [1.38]	5.92 [1.15]	18.32 [3.67]	9.74 [1.92]	7.95 [1.41]	10.26 [2.02]	4.01 [0.68]
umd	1.45 [0.86]	1.13 [0.66]	2.66 [1.60]	2.20 [1.29]	2.41 [1.39]	2.57 [1.51]	0.47 [0.28]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	12	10	11	8	7	8	15

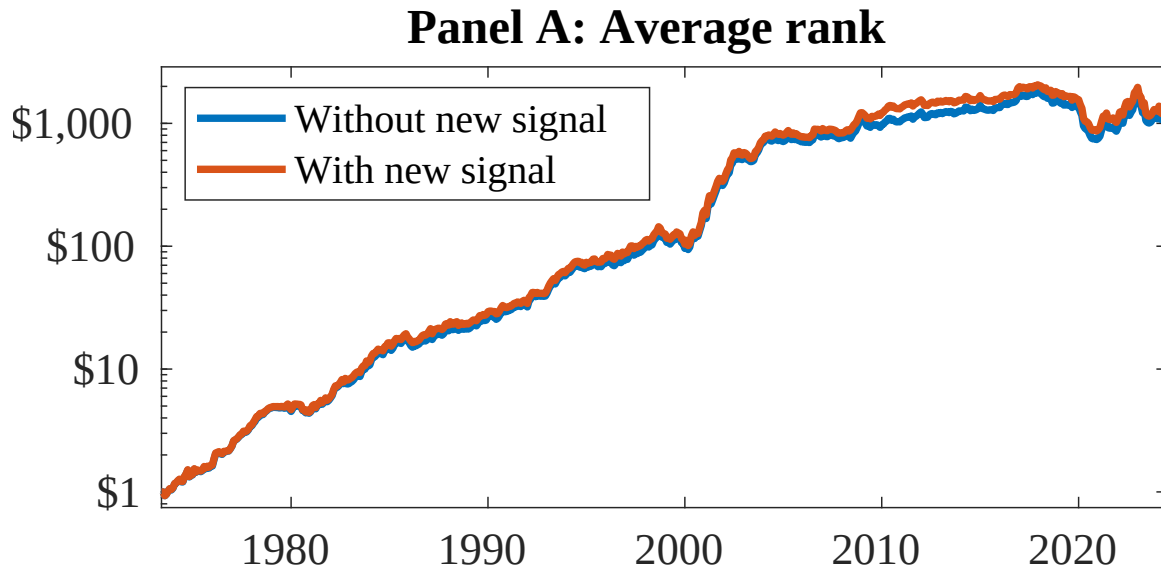


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as ELS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. (2004). Long-run risks and financial markets. *Federal Reserve Bank of St. Louis Review*, 89(4):1–17.
- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H., Piazzesi, M., and Tuzel, S. (2023). Consumption-based asset pricing: Evidence from new tests. *Journal of Financial Economics*, 148(2):273–296.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- George, T. J. and Hwang, C.-Y. (2010). A resolution of the distress risk and leverage puzzles in the cross section of stock returns. *Journal of Financial Economics*, 96(1):56–79.
- Gomes, J. a. F. and Schmid, L. (2010). Levered returns. *Journal of Finance*, 65(2):467–494.
- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Sloan, R. G. (1996). Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3):289–315.
- Spiess, D. K. and Affleck-Graves, J. (1999). The long-run performance of stock returns following debt offerings. *Journal of Financial Economics*, 54(1):45–73.
- Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.